

DAIVIK BHATIA (DTGS)

Graphics Driver Developer (Mesa)

PROFESSIONAL SUMMARY

Software Developer with a strong foundation in systems programming and hardware acceleration, transitioning from AI/ML engineering to Graphics Driver Development. Active contributor to the Mesa 3D project with a focus on Broadcom (Raspberry Pi) drivers.

FOSS CONTRIBUTIONS

Mesa, 01/2025 to Current

- As a contributor to **Mesa**, I work on improving the **v3d** and **v3dv drivers** for Broadcom hardware. I'm currently adding new **Vulkan features** like robustness2 to make the driver more stable and reliable.
- A big part of my role involves debugging and fixing active bugs, as well as cleaning up the codebase by resolving "FIXME" tasks to make the driver logic more complete and efficient.

Pyboy Emulator, 07/2024

- I contributed to **PyBoy** by centralizing the **frame limiter** logic, moving it into a base plugin so it works across all window types, including headless "Null" windows. I also built out a **GameShark API**, creating a flexible system that allows users to load and run cheats either directly from strings or by importing external text files.

WORK HISTORY

AI Engineer III, 08/2022 to Current

American Express - Gurgaon, India

- I developed a **NLP API** to automate policy-based customer queries using **LLMs**, built a **machine learning framework** for high accuracy entity matching between internal and third party data, and led the creation of a new **data hierarchy system**.

ML Engineer, 01/2022 to 08/2022

Koo App - Bengaluru, India

- I designed and deployed a suite of **REST APIs** for automated image intelligence, specializing in **OCR text extraction**, **NSFW content filtering**, and **AI-driven image captioning**.

R&D Engineer (AI/ML), 07/2021 to 01/2022

Logic Fruit Technologies - Gurgaon, India

- I competed in the **DAC SDC 2021 research challenge**, where I built a high-speed object detection system for the **Avnet Ultra96 v2 FPGA**. I experimented with several models like **YOLO** and **EfficientDet**, optimizing them to hit a peak performance of 72 FPS.

CONTACT

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PORTFOLIOS

- <https://gitlab.freedesktop.org/dtgs1208>
- <https://daivikbhatia.github.io/portfolio-live/>

SKILLS

- **Graphics & Systems:** C, C++, Vulkan, OpenGL, GLSL
- **Development Tools:** Git, GDB, KGDB, VK-GL-CTS, Piglit
- **General Programming:** Python, SQL, Java script

EDUCATION

B.Tech in Electronics & Communication Engineering,
06/2021

ADGIPS, GGSIPU - Delhi, India

LANGUAGES

English

Hindi